

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Hospital discharge summaries contain essential details about a patient's diagnosis, treatment, medications, and follow-up instructions. However, these documents are often highly technical and difficult for patients to interpret without medical knowledge. Additionally, patients face challenges such as understanding dietary restrictions, verifying billing accuracy, and finding affordable sources for prescribed medicines. Traditional systems do not provide automated assistance that combines medical interpretation, financial verification, and cost-saving recommendations in a single platform. With advancements in AI, multi-agent systems offer a powerful solution for improving how patients interact with and understand healthcare information. This project introduces an intelligent, automated multi-agent architecture that helps patients navigate post-discharge information more easily and confidently.

1.2 Problem Statement

Patients commonly struggle to interpret hospital discharge summaries, understand prescribed diet requirements, verify the accuracy of hospital bills, and locate affordable sources for medications. Existing systems provide no automated support to simplify medical terminology, validate billing charges, or compare medicine prices. As a result, patients experience confusion, financial uncertainty, and limited guidance during a critical recovery period. Therefore, there is a need for an intelligent, automated system that can process discharge documents, retrieve relevant information, validate billing accuracy, generate personalized recommendations, and assist patients through an easy-to-use chatbot interface.

1.3 Objectives

The primary objective of this project is to design and implement a multi-agent AI system that enhances patient understanding and supports post-discharge decision-making. Specific objectives include:

- To simplify complex discharge documents into clear, patient-friendly summaries.
- To generate personalized diet recommendations based on the patient's medical condition.
- To validate hospital bills by detecting pricing errors and policy violations using regulatory data.
- To compare medicine prices across online pharmacies and suggest cost-effective options.
- To integrate RAG-based agents and API-driven agents under a coordinated Orchestrator Agent.
- To deliver accurate, reliable, and easy-to-understand responses through a chatbot interface.

1.4 Existing System

In existing setups, hospitals provide patients with discharge documents in printed or PDF form, leaving them responsible for interpreting the medical terminology, dietary instructions, and billing details on their own. Patients often depend solely on doctors or hospital staff for clarification, which may not always be accessible. Bill verification is typically manual, making it difficult for patients to identify overcharging or policy mismatches. Similarly, comparing medicine prices across online pharmacies must be done manually, which is time-consuming and prone to errors. Current systems lack automation, personalization, transparency, and an integrated platform that addresses all post-discharge needs in one place.

1.5 Proposed System

The proposed system introduces a comprehensive Multi-Agent AI architecture that automates the interpretation and processing of discharge documents. After the patient uploads the PDF, the system uses OCR and preprocessing techniques to extract structured information and store it in a Vector Store for semantic retrieval. A chatbot interface allows patients to ask questions, which are processed by an Orchestrator Agent. The orchestrator intelligently routes the request to specialized agents: the Discharge Summary Agent simplifies medical content; the Diet & Nutrition Agent generates personalized meal plans; the Bill Validator Agent checks charges against NPPA and insurance rules; and the Medicine Price Comparison Agent retrieves real time prices from online pharmacies through API calls. The orchestrator then compiles these outputs into a unified, user-friendly response. This system ensures accuracy, transparency, and convenience while reducing the patient's effort in understanding post-discharge information.

CHAPTER 2: LITERATURE SURVEY

Relevant Research Papers and Reports The following academic materials and reports have been compiled to provide reliable support for your project:

- **Category: Healthcare AI & Patient Education**

Title: “AI-generated patient-friendly discharge summaries to empower patients”

Authors: Nils Reuter, Vincent-Noah von Lipinski, Jonathan Jeutner, Thorsten Schlomm, Martin Witzenrath, Leif Erik Sander, Matthias Gröschel

Relevance: Investigates the utility of Large Language Models (specifically GPT-4o) in creating simplified discharge summaries to improve health literacy. The study finds that 90% of patients reported better understanding of their condition after reading the AI summary, often rating it as more helpful than standard physician discharge consultations.

- **Category: Healthcare AI & Clinical Documentation**

Title: “Abstract Meaning Representation for Hospital Discharge Summarization”

Authors: Paul Landes, Sitara Rao, Aaron J. Chaise, and Barbara Di Eugenio

Relevance: Proposes a graph-based, extractive method to generate hospital discharge summaries from clinical notes using Abstract Meaning Representation (AMR). The study specifically targets the "hallucination" problem in Large Language Models (LLMs) by ensuring that every part of the generated summary is traceable back to the original patient records, making it safer for clinical use than standard generative.

- **Category: AI in Nutrition & Personalized Health**

Title: “AI nutrition recommendation using a deep generative model and ChatGPT”

Authors: Ilias Papastratis, Dimitrios Konstantinidis, Petros Daras & Kosmas Dimitropoulos

Relevance: Proposes a hybrid framework that combines a deep generative model (Variational Autoencoder) for precise user profiling with ChatGPT to expand meal variety. The study demonstrates that this method eliminates the caloric and macronutrient inaccuracies often found in standalone Large Language Model recommendations by using an optimizer and strict nutritional loss functions to align with EFSA/WHO guidelines.

- **Category: Healthcare AI & Insurance Fraud Detection**

Title: “Medical Insurance Claim Validation System”

Authors: Jitesh Shringare, Shruti Sabat, Lalit Patil, Sagrika Waghmare, and Raju Mendhe

Relevance: This paper presents the **Medical Insurance Claims Verification System (MICVS)**, a framework designed to automate and streamline the insurance claim approval process using machine learning and big data analytics.

- **Category: Pharmacoeconomics & Healthcare Policy**

Title: “A Comparative Study of Branded and Generic Drugs: Cost Analysis in Indian Scenario”

Authors: Prasad T A, Panesar A, and Suttee A

Relevance: This study conducts a comprehensive pharmacoeconomic analysis to validate the cost-effectiveness of unbranded generic medicines (specifically from the *Jan Aushadhi Scheme*) compared to branded alternatives in the Indian market.

CHAPTER 3: METHODOLOGY

The methodology describes the step-by-step process followed to build an intelligent multi-agent healthcare assistance system for post-discharge support. The system follows a structured pipeline that begins with document ingestion and text extraction, followed by semantic processing, agent-based reasoning, and final response generation. The methodology integrates OCR, preprocessing, embedding generation, vector-based retrieval, multi-agent coordination, and API-based data fetching to ensure accurate, reliable, and patient-friendly outputs.

1. User Upload of Discharge PDF

The methodology begins with user interaction through a hospital portal chatbot. Patients upload their hospital discharge summaries in PDF format, which may include digitally generated documents or scanned copies. The system is designed to support heterogeneous document formats, ensuring inclusivity for users regardless of how their discharge summary was issued. This step serves as the primary data entry point and ensures secure handling of sensitive medical documents through authenticated access and encrypted transmission.

2. Optical Character Recognition (OCR) Extraction

Once the discharge PDF is uploaded, it is passed to an OCR module responsible for extracting readable text. For scanned or image-based PDFs, OCR technology converts visual content into machine-readable text. This step is critical in healthcare environments where many discharge summaries are still issued as scanned documents. The OCR module is optimized to handle medical terminology, abbreviations, and tabular data such as medication lists and billing details. The extracted text is stored temporarily for further processing.

3. Text Preprocessing and Chunking

Once the raw text is extracted, preprocessing is performed to improve data quality and usability. This includes removing noise, correcting formatting issues, and standardizing medical text. Following preprocessing, the cleaned text is divided into smaller, semantically meaningful chunks. Chunking improves retrieval efficiency and ensures that relevant information can be accessed precisely during query handling. Each chunk is designed to maintain contextual integrity while remaining within the token limits required for embedding generation and retrieval.

4. Embedding Generation

Each text chunk generated in the previous stage is converted into a numerical representation known as an embedding vector. Embeddings capture the semantic meaning of text, allowing the system to perform similarity-based retrieval rather than simple keyword matching. Advanced embedding models are employed to ensure that medical context, clinical terminology, and patient-specific information are accurately represented. These embeddings form the foundation for semantic search and Retrieval-Augmented Generation (RAG) within the system.

5. Indexing into Vector Store

All generated embeddings are indexed and stored in a Vector Store, which serves as the system's primary knowledge base. In addition to discharge document embeddings, the Vector Store also contains external reference documents, including National Pharmaceutical Pricing Authority (NPPA) drug pricing data, government healthcare regulations, insurance policy clauses, and standardized dietary guidelines. This integrated knowledge base enables comprehensive retrieval across both patient-specific and authoritative external data sources, thereby enhancing response accuracy and reliability.

6. User Query Interaction via Chatbot

After document processing is complete, the patient interacts with the chatbot interface to ask questions related to their discharge summary. Queries may include requests for simplified explanations of diagnoses, personalized diet recommendations, verification of hospital bills, or comparison of medicine prices. The chatbot acts as a conversational interface, capturing user input and forwarding it to the Orchestrator Agent for further analysis. This design ensures a user-friendly experience while abstracting system complexity from the patient.

7. Intent Detection by Orchestrator Agent

The Orchestrator Agent plays a central role in the multi-agent architecture. Upon receiving a user query, it performs intent detection to understand the nature and objective of the request. Using NLP-based classification techniques, the Orchestrator determines whether the query pertains to discharge summarization, dietary guidance, billing validation, or medicine price comparison. Based on this analysis, the Orchestrator dynamically assigns the task to the most appropriate specialized agent, ensuring efficient and accurate handling of diverse user needs.

8. Task Execution by Specialized Agents

Specialized agents are responsible for executing domain-specific tasks. The Discharge Summary Agent retrieves relevant discharge-related chunks from the Vector Store and generates simplified, patient-friendly explanations using RAG techniques. The Diet and Nutrition Agent combines patient condition data with dietary guidelines to produce personalized diet plans. The Bill Validator Agent cross-references hospital billing details with insurance clauses and regulatory norms to identify discrepancies or overcharges.

In contrast, the Medicine Price Comparison Agent performs real-time API calls to online pharmacy platforms such as 1mg and PharmEasy. This agent retrieves up-to-date medicine pricing information, enabling patients to compare costs and make informed purchasing decisions. The combination of static RAG-based agents and dynamic API-driven agents ensures both contextual depth and real-time accuracy.

9. Return of Agent Outputs to Orchestrator

After task completion, each specialized agent returns its generated output to the Orchestrator Agent. These outputs may vary in structure and content depending on the task performed.

The Orchestrator ensures that all responses are validated for consistency, relevance, and factual correctness before proceeding to the final stage. This step reinforces the reliability of the system by preventing fragmented or conflicting information from reaching the user.

10. Aggregation and Formatting of Final Response

The Orchestrator Agent aggregates the outputs received from multiple agents into a unified response. The information is structured logically, formatted clearly, and simplified to ensure patient comprehension. Medical jargon is minimized, key points are highlighted, and explanations are tailored to a non-technical audience. This aggregation step transforms complex clinical and administrative data into actionable insights for patients.

11. Response Delivery to Chatbot Interface

In the final stage, the Orchestrator sends the formatted response back to the chatbot interface. The patient receives clear and concise information such as a simplified discharge summary, personalized diet recommendations, validated billing details, or comparative medicine pricing. This completes the end-to-end methodology, ensuring that patients are empowered with understandable, accurate, and timely healthcare information through an intelligent multi-agent system.

The proposed methodology demonstrates a robust and scalable approach for healthcare document processing using modern AI techniques. By combining OCR, semantic retrieval, multi-agent coordination, and real-time data integration, the system effectively bridges the gap between complex medical documentation and patient-friendly communication.

CHAPTER 4: SYSTEM DESIGN

4.1 Overview

The system design follows a modular, multi-agent architecture that processes patient discharge documents, extracts relevant information, and provides personalized responses through a hospital chatbot. The architecture combines OCR, preprocessing, RAG (Retrieval-Augmented Generation), LLM-based reasoning, vector database retrieval, and external API integration. The aim is to deliver an intelligent, automated, and user-friendly healthcare assistance system.

4.2 System Architecture

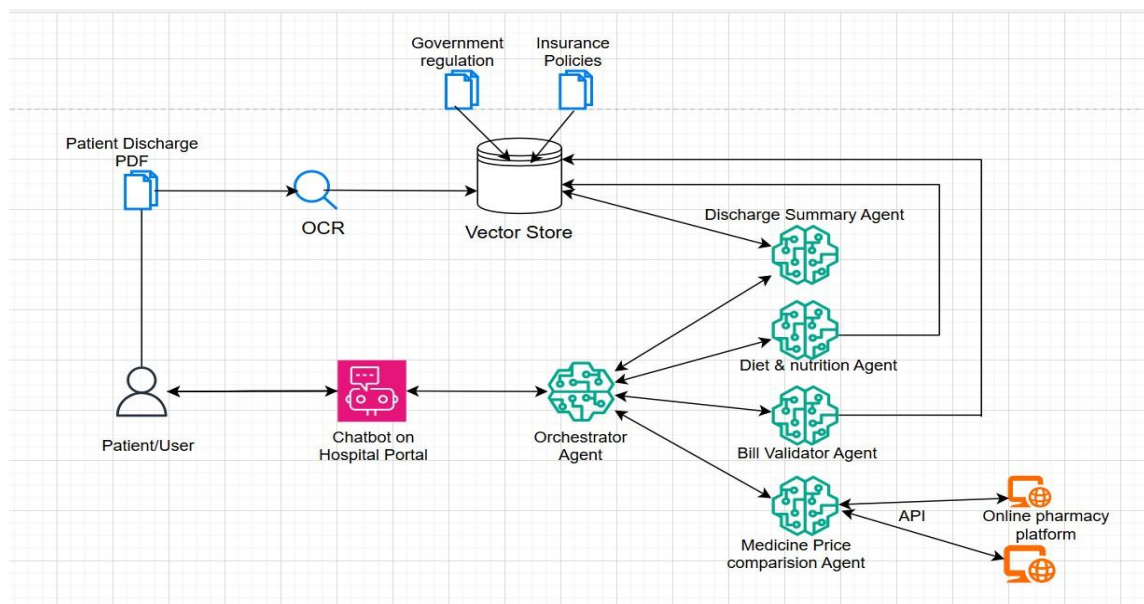


Figure 3.1: System Architecture of the Multi-Agent Healthcare AI System

Step 1: Discharge PDF → OCR

The user uploads the discharge PDF through the hospital portal chatbot. The OCR module extracts readable text from scanned or image-based documents.

Step 2: Preprocessed Text → Vector Store

The extracted text undergoes preprocessing, chunking, and embedding generation. These embeddings, along with supporting documents such as government regulations, insurance policies, and NPPA pricing data, are stored in the Vector Store.

Step 3: User Query via Chatbot

The patient interacts with the chatbot and asks questions such as:

- “Explain my discharge summary.”
- “Give me a diet plan.”
- “Check if my bill has overcharging.”
- “Find the cheapest price for my medicines.”

The chatbot forwards the query to the Orchestrator Agent.

4.3 Multi-Agent Architecture (Component Descriptions)

AGENT 1: ORCHESTRATOR AGENT

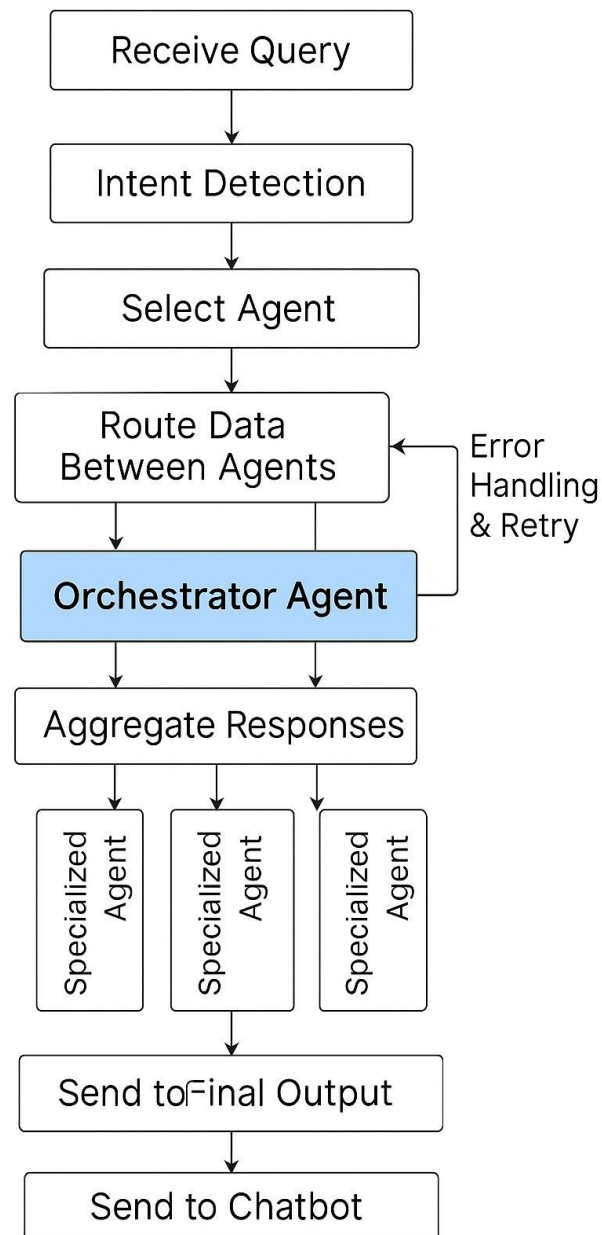


Figure 4.3.1: Orchestrator Agent Workflow

The Orchestrator Agent functions as the central controller of the system. Its key roles include understanding the user's intent from the chatbot query, selecting the appropriate specialized agent to handle the request, routing data between agents, invoking multiple agents in parallel when required, aggregating the outputs produced by each agent into a unified final response, and formatting the information in a readable and user-friendly manner. In addition to this, the orchestrator manages error handling, implements retry mechanisms, and ensures smooth communication across the entire multi-agent workflow.

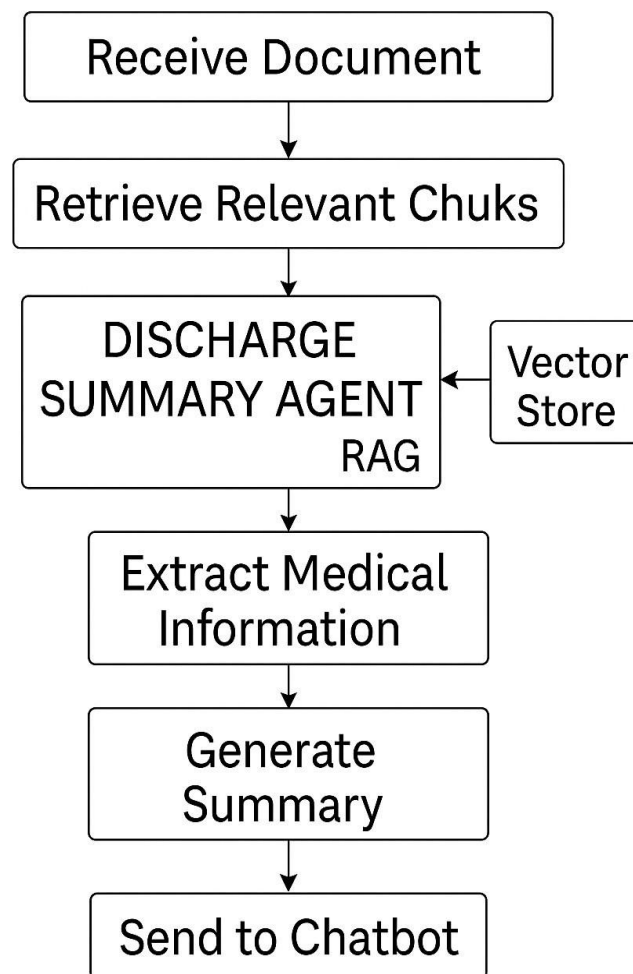
AGENT 2: DISCHARGE SUMMARY AGENT**Implementation Stack**

Figure 4.3.2: Discharge Summary Agent Workflow

The Discharge Summary Agent uses Retrieval-Augmented Generation (RAG) to produce an easy-to-understand summary of the patient's discharge document. It retrieves relevant text chunks from the Vector Store, extracts important medical information such as diagnosis, procedures, medications, and laboratory findings, and uses a large language model to convert complex medical terms into simple, patient-friendly explanations. The agent ensures that the summary maintains clinical accuracy while making the information more understandable for patients.

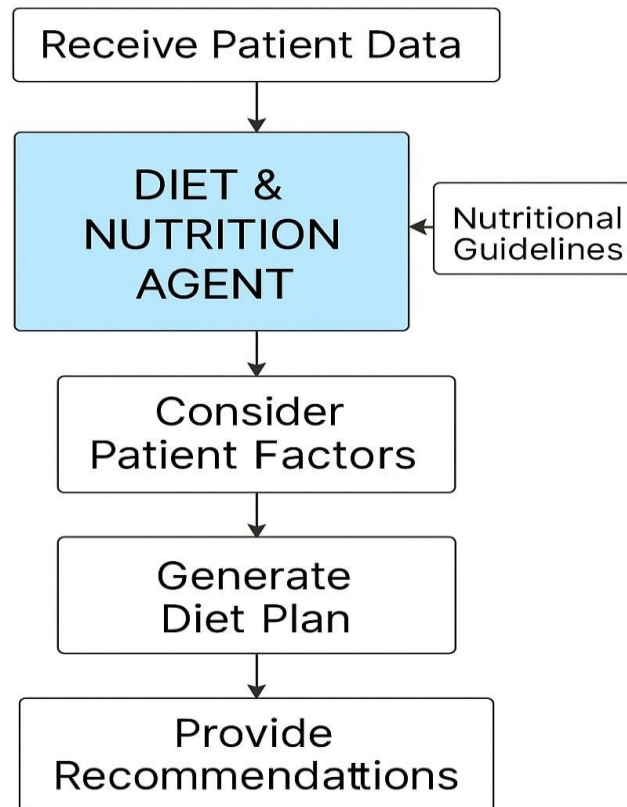
AGENT 3: DIET & NUTRITION AGENT

Figure 4.3.3: Diet & Nutrition Agent Workflow

The Diet & Nutrition Agent operates using a hybrid RAG and rule-based approach to generate personalized dietary recommendations for patients. It retrieves disease-specific nutritional guidelines, considers the patient's medications, allergies, lifestyle, and cultural food preferences, and produces structured diet plans that include appropriate Indian meal suggestions, foods to include, foods to avoid, and daily or weekly nutritional charts. This agent helps patients understand what they should eat during recovery and promotes overall health management.

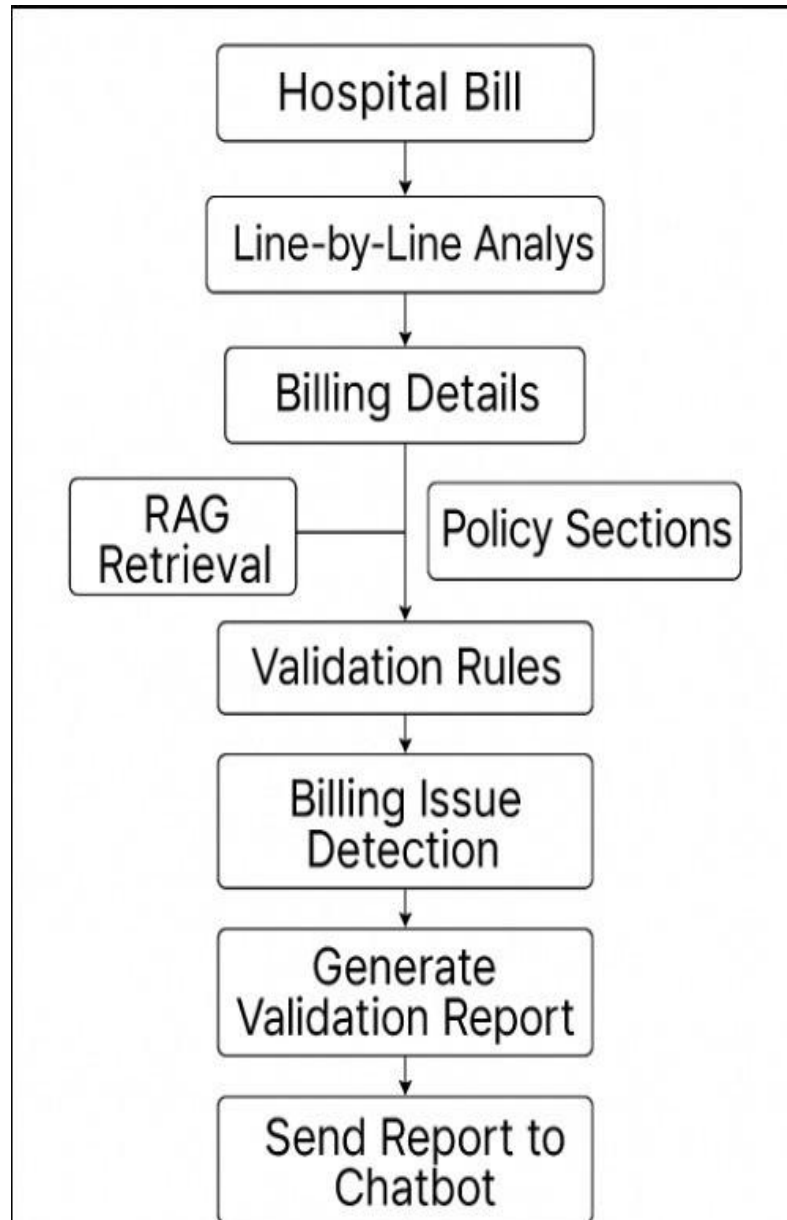
AGENT 4: BILL VALIDATOR AGENT

Figure 4.3.4: Bill Validator Agent Workflow

The Bill Validator Agent is a RAG-based auditing system that scrutinizes hospital bills line-by-line to identify overcharging, duplicate billing, and billing errors. It validates medicine prices against NPPA (National Pharmaceutical Pricing Authority) ceiling prices, and checks insurance policy terms for room rent limits and co-payment clauses. The agent uses RAG to retrieve relevant policy sections from the Vector Store and applies validation rules to detect common billing issues like unbundling, upcoding, and non-medical expenses. It generates a detailed validation report highlighting discrepancies with evidence, calculates potential savings, and provides negotiation talking points for patients to contest incorrect charges.

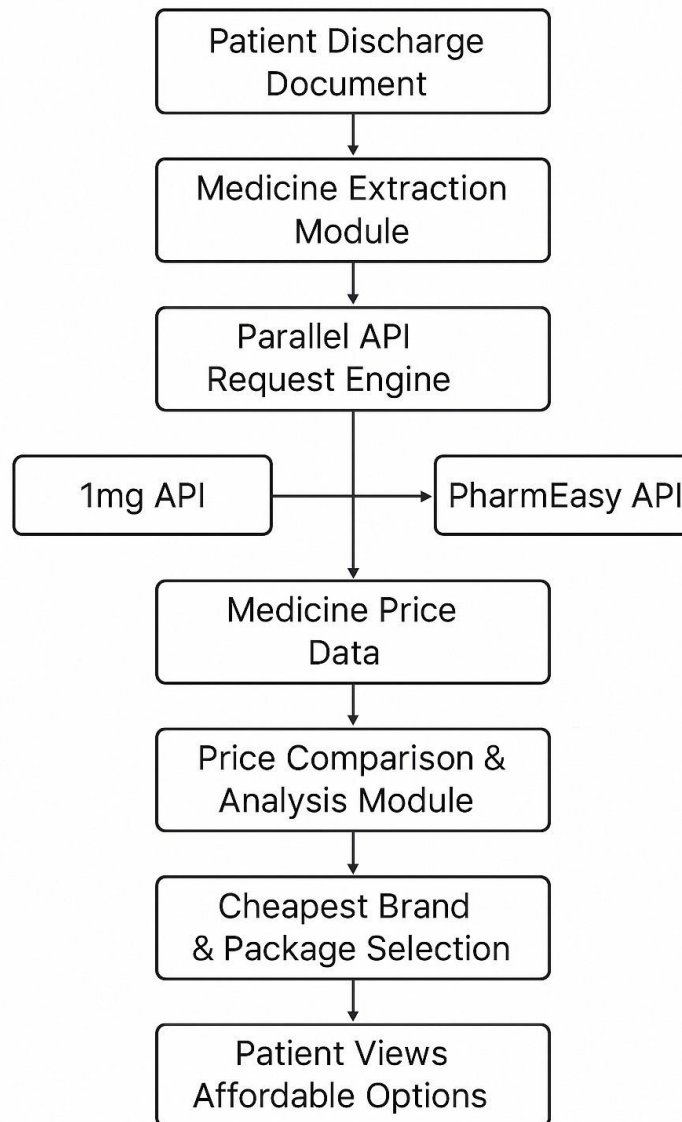
AGENT 5: MEDICINE PRICE COMPARISON AGENT

Figure 4.3.5: Medicine Price Comparison Workflow

The Medicine Price Comparison Agent relies on external APIs rather than RAG. It extracts the list of prescribed medicines from the discharge information, performs parallel API calls to online pharmacy platforms such as 1mg and PharmEasy, retrieves real-time prices for various brands, dosages, and package sizes, compares the results, and identifies the cheapest purchasing options available. This enables patients to select affordable sources for their medicines and reduce their expenses.

4.4 Final Output Delivery

After each specialized agent completes its task, the results are sent back to the Orchestrator Agent, which combines all outputs into a single, clear, and well-structured response. The orchestrator formats the final answer for readability and sends it to the chatbot interface. The patient then receives an organized summary that may include a simplified discharge explanation, a personalized diet plan, a bill validation overview, and affordable medicine price suggestions.

CHAPTER 5: TECHNOLOGY STACK AND IMPLEMENTATION PLAN

5.1 Technology Stack

The development of the Intelligent Multi-Agent Healthcare Assistant utilizes a combination of modern AI frameworks, data-processing tools, and backend technologies to ensure efficient document handling, semantic retrieval, multi-agent coordination, and interactive user communication. The major components of the technology stack are:

1. Programming Languages

- **Python:** Primary language for building agents, integrating OCR, generating embeddings, and implementing Retrieval-Augmented Generation (RAG).

2. AI & NLP Technologies

- **Large Language Models (LLMs):** Used for summarization, reasoning, explanation generation, and natural language query handling.
- **RAG (Retrieval-Augmented Generation):** Enhances accuracy by retrieving relevant information from the Vector Store.
- **Embeddings Models:** Used to convert discharge content and reference documents into vector representations for semantic search.

3. OCR & Text Processing

- **Tesseract OCR / EasyOCR:** Extracts readable text from scanned or image-based discharge PDFs.
- **Text cleaning and preprocessing tools:** Tokenization, normalization, chunking, noise removal.

4. Vector Database

- **FAISS / Pinecone / ChromaDB:** Stores embeddings and enables high-speed semantic retrieval for agents such as Discharge Summary, Diet & Nutrition, and Bill Validator.

5. Multi-Agent Framework

- **LangChain Agents / Custom Agentic Framework:** Used to create the Orchestrator Agent and specialized functional agents.
- **Tool calling & routing mechanisms:** Enable interaction among agents for coordinated execution.

6. API Integrations

- **Online pharmacy APIs (1mg, PharmEasy):** Used by the Medicine Price Comparison Agent for real-time price retrieval.
- **Insurance policy and NPPA dataset integrations:** Used by the Bill Validator Agent for compliance checks.

7. Backend & Deployment

- **Flask / FastAPI:** Backend for handling chatbot queries and agent communication.
- **Docker (optional):** Containerization for scalability and easy deployment.
- **Cloud Hosting (AWS/GCP/Azure):** For hosting the model, vector store, and agents.

5.2 Implementation Plan

The implementation follows a structured development process, ensuring modular and scalable system construction.

Phase	Duration	Key Activities & Milestones
1. Foundation Setup	2 Weeks	Set up Git repository, project structure, virtual environment, and install required AI, OCR, and vector database libraries. Configure the OCR pipeline and initialize backend skeleton using Flask/FastAPI. Milestone: Project environment ready; OCR module functioning.
2. Document Processing & Vector Store Setup	3 Weeks	Implement OCR-to-text pipeline, perform text preprocessing (cleaning, chunking), generate embeddings, and configure Vector Store (FAISS/Pinecone/ChromaDB). Upload NPPA rules, insurance policies, and dietary guidelines.
3. Agent Development	5 Weeks	Develop and test core agents: Discharge Summary Agent (RAG), Diet & Nutrition Agent (RAG + rule-based), Bill Validator Agent (RAG + rules), and Medicine Price Comparison Agent (API). Each agent operates independently and returns structured outputs. Milestone: All agents operational and testable.
4. Orchestrator & Workflow Integration	3 Weeks	Implement Orchestrator Agent with intent detection, agent routing, and output aggregation. Integrate all agents into a seamless workflow and ensure inter-agent communication. Milestone: Fully functional backend with working multi-agent pipeline.
5. Chatbot Interface & User Interaction	2 Weeks	Develop the chatbot interface and connect it to backend APIs. Implement user query handling, formatting of responses, and conversation flow. Milestone: Chatbot capable of complete end-to-end interaction.
6. Testing, Validation & Optimization	3 Weeks	Conduct end-to-end testing for all agents, evaluate RAG retrieval accuracy, API stability, bill validation correctness, and diet plan precision. Optimize performance, error handling, and system reliability.
7. Final Deployment & Demonstration	2 Weeks	Prepare cloud/Docker deployment of the system. Finalize UI, prepare demonstration dashboard, and complete project documentation for viva. Milestone: Production-ready multi-agent healthcare assistant with live demo capability.

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CONCLUSION

This document has presented the complete design of an *Intelligent Multi-Agent Healthcare Assistant for Post-Discharge Support*. The proposed system integrates OCR, RAG-based retrieval, vector storage, and multiple specialized agents coordinated through an intelligent Orchestrator. This architecture enables automated extraction, interpretation, and presentation of discharge information, diet recommendations, bill validation insights, and real-time medicine price comparisons.

The design clearly defines the functionality of each agent, the shared Vector Store that supports semantic understanding, and the overall workflow that ensures efficient inter-agent communication. By simplifying complex medical and financial information, the system aims to improve patient clarity and enhance post-discharge decision-making.

The next stage of this project will involve implementing the designed architecture, followed by systematic testing and validation. Successful completion will provide a functional and reliable AI-based healthcare assistant, serving as a strong foundation for future enhancements and broader healthcare applications.